NR241 / NR242 / NR210 — Job File Guide

This guide describes the job file used by **NR241, NR242, NR210 and variants using this format with the MSC interface**. Field definitions come from `NR24X- job file description.xlsx`. Each data row represents one part to be cut; rows with the same bar number represent parts produced from the same bar.

These machines use `nccode`: it carries machining information such as holes, drainage and slots. The “macros are not used” guidance for TT machines with the HMS_W interface does not apply to the NR machines covered here.

File fields

The reference document contains **30 columns**:

```
id;ksn;ksnbar;ktn;ktnbar;l;r;code;info;width;height;trolley;box;orientation;reinf;reinfbar;pos;prono;offno;ag
ent;customer;date;nccode;isfix;colorcode;colorinfo;mainprofile;subcust;image;DATA1
```

This illustration uses ; as the separator. Match the separator and numeric format to the machine's import settings. The Excel document describes the fields; it does not define every machine's import configuration. When a value is empty, preserve its column position rather than shifting later values to the left.

Fields marked **Machine** below are marked with > as related to machine operation in the source document. This is not a list of mandatory columns. **Label / extra information** fields describe order tracking and label content; their use depends on the machine and label configuration.

Field	Use	Description
`id`	Machine	Part identifier; increases through the job file.
`ksn`	Machine	Bar number. Rows sharing the same value are cut from the same bar.
`ksnbar`	Machine	Bar length. The last digit is the decimal place: `59800` = **5980.0 mm**.
`ktn`	Machine	Part sequence within the same bar.
`ktnbar`	Machine	Cut part length. The last digit is the decimal place: `4710` = **471.0 mm**.
`l`	Machine	Cutting angle at the start of the part; `45` in the source example.
`r`	Machine	Cutting angle at the end of the part; `135` in the source example.
`code`	Machine	Profile / bar code for the part.
`info`	Label / extra information	Profile or bar description.
`width`	Label / extra information	Window / frame width.
`height`	Label / extra information	Window / frame height.
`trolley`	Label / extra information	Trolley number for the part.
`box`	Label / extra information	Box / compartment number on the trolley.
`orientation`	Label / extra information	Part location in the frame, such as top, bottom, right or left.
`reinf`	Label / extra information	Reinforcement code.

Field	Use	Description
`reinfbar`	Label / extra information	Reinforcement length.
`pos`	Label / extra information	Window / position number; parts of the same window share the value.
`prono`	Label / extra information	Production number.
`offno`	Label / extra information	Contract number.
`agent`	Label / extra information	Dealer / distributor information.
`customer`	Label / extra information	Customer information.
`date`	Label / extra information	Date information; check its representation with the system creating the file.
`nccode`	**Machine**	**Machining / macro information.** One or more 18-character operation blocks.
`isfix`	Label / extra information	The source example contains `0`; its function is not separately explained.
`colorcode`	Label / extra information	Colour code.
`colorinfo`	Label / extra information	Colour description.
`mainprofile`	Machine	Main profile / bar code.
`subcust`	Label / extra information	Additional customer information.
`image`	Label / extra information	Path to the image printed on the label.
`DATA1`	Label / extra information	Additional information that may be printed on the label.

`agent` is a separate column. Do not confuse this 30-column file with other SC examples that omit it. The Excel document does not provide a detailed colour-code table, so another machine's colour mapping should not be applied directly.

Bar and part sequence

id	ksn	ktn	Meaning
1	1	1	First part of the first bar
2	1	2	Second part of the first bar
3	2	1	First part of the second bar

`id` identifies the part across the file. `ktn` identifies its sequence within a bar. In the reference sample, bar 1 contains four parts and bar 2 contains five parts.

Length and angle notation

- `ksnbar = 59800` → **5980.0 mm** bar length.
- `ktnbar = 4710` → **471.0 mm** part length.
- `l = 45`, `r = 135` → start and end cutting angles.

The source defines the final digit as the decimal place for `ksnbar` and `ktnbar`. Do not automatically extend this rule to `width`, `height` or `reinfbar`. The angle examples here are single numbers; do not replace them with the `45|90` notation from HMS_W SC examples.

`nccode` — machining / macro field

Each operation is an **18-character** block. For multiple operations on the same part, concatenate the blocks.

```
H | 001 | 02 | 03 | 05000 | 01000
```

Vertical bars are explanatory and are not part of the actual field.

Section	Character positions	Length	Meaning
`H`	1	1	Fixed macro start character
`001`	2–4	3	Tool code
`02`	5–6	2	Operation code
`03`	7–8	2	Accessory code
`05000`	9–13	5	Operation position: **500.0 mm**
`01000`	14–18	5	Slot length: **100.0 mm**

The first 8 characters, **`H0010203`**, mean `H` + Tool `001` + Operation `02` + Accessory `03`. Preserve leading zeros. The full 18-character example is:

```
H00102030500001000
```

Only **`H`** is the fixed start character. The following `0` belongs to the three-digit Tool code. The remaining 10 digits contain the five-digit position and five-digit slot length.

Drainage / slot example

```
H00201010500001000
```

This example uses Tool **`002`**, Operation **`01`** and Accessory **`01`**, with position **500.0 mm** and slot length **100.0 mm**. For drainage or a slot, the position indicates the **start of the operation**.

Hole example

```
H01601011000000000
```

This example uses Tool **`016`**, Operation **`01`** and Accessory **`01`**, with position **1000.0 mm**. The last five digits are `00000`, so there is no slot length. In this non-drainage / non-slot example, the position indicates the **hole centre**.

Multiple operations on the same part

```
H00201010500001000H01601011000000000
```

This field contains two 18-character blocks, **36 characters** in total. The first describes drainage / a slot; the second describes a hole. Interpret the position according to the operation type.

These codes are examples from the reference document, not a universal operation list for every profile and machine. Check the code's machine operation against Macro Mapping and the profile's operation definitions. Correct cutting lengths alone are not enough: an empty or incorrect `nccode` can leave required operations missing.

Illustrative file

Download the 30-column example CSV

Dealer, customer, order, profile and image-path values are fictitious. The separator is ;. This example illustrates the fields; match it to your own machine's profile, machining and import settings before production.

Checks before import

1. Confirm that the file uses the NR job-file layout prepared for your machine.
2. Compare the headers and field count in every row; empty fields must not shift later values.
3. Check bar numbers, part sequence within each bar, bar / part lengths and cutting angles.
4. Verify that profile and main-profile codes match the correct machine definitions.
5. Check the 18-character `nccode` blocks, Tool / Operation / Accessory codes, positions and slot lengths where applicable. Preserve leading zeros.
6. After import, review the part lengths and machining display.

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